

1. Mastering the translation between tensor language and component expressions

Solving these problems provides you with all key skills for tensor calculations.

Let $\varphi: V \rightarrow V$ be a linear map on a vector space $(V, \oplus, \odot)$ and $e_1, \dots, e_{\dim V}$ a basis for V .

(a) Prove that a linear map $\varphi: V \rightarrow V$ defines a $(1, 1)$ -tensor

$$T_\varphi: V^* \times V \rightarrow \mathbb{R}, \quad T(\omega, v) := \omega(\varphi(v)).$$

Solution: We need to check linearity of T_φ in each slot. For the first slot, one has for all $\omega, \omega' \in V^*$, $v \in V$ and $\lambda \in \mathbb{R}$ that

$$T_\varphi(\omega \boxplus \lambda \boxminus \omega', v) = (\omega \boxplus \lambda \boxminus \omega')(\varphi(v)) = \omega(\varphi(v)) + (\lambda \boxminus \omega')(\varphi(v)) = T_\varphi(\omega, v) + \lambda T_\varphi(\omega', v),$$

where we used the ‘point-wise defined’ operations $\boxplus$ and $\boxminus$ on V^* . Note that this one calculation shows additivity (by setting $\lambda = 1$) and the scaling property (by letting $\omega = O_{V^*}$) for the first slot. Similarly, for the second slot, one has for all $\omega \in V^*$, $v, v' \in V$ and $\lambda \in \mathbb{R}$ that

$$T_\varphi(\omega, v \oplus \lambda \odot v') = \omega(\varphi(v \oplus \lambda \odot v')) = \omega(\varphi(v) \oplus \lambda \odot \varphi(v')) = \omega(\varphi(v)) + \lambda \omega(\varphi(v')) = T_\varphi(\omega, v) + \lambda T_\varphi(\omega, v'),$$

which shows linearity in the second slot. Hence $T_\varphi: V^* \times V \rightarrow \mathbb{R}$ is a $(1, 1)$ -tensor.

(b) Show that defining the components φ^i_j of the linear map φ as the expansion coefficients for the image $\varphi(e_j)$ of the j -th basis vector e_j in terms of the same basis in the target space, namely

$$\varphi(e_j) = \bigoplus_{i=1}^{\dim V} \varphi^i_j \odot e_i,$$

one has

$$\varphi^i_j = (T_\varphi)^i_j,$$

where $(T_\varphi)^i_j$ are the components of the tensor T_φ .

Solution: The components of the $(1, 1)$ -tensor T_φ are defined in terms of the basis $e_1, \dots, e_{\dim V}$ and the dual basis $\epsilon^1, \dots, \epsilon^{\dim V}$ as

$$(T_\varphi)^i_j := T_\varphi(\epsilon^i, e_j) = \epsilon^i(\varphi(e_j)) = \epsilon^i\left(\bigoplus_{k=1}^{\dim V} \varphi^k_j \odot e_k\right) = \sum_{k=1}^{\dim V} \epsilon^i(\varphi^k_j \odot e_k) = \sum_{k=1}^{\dim V} \varphi^k_j \underbrace{\epsilon^i(e_k)}_{\delta_k^i} = \varphi^i_j,$$

where the linearity of the map $\varphi: V \rightarrow V$ and of the maps $\epsilon^i: V \rightarrow \mathbb{R}$ has been used, besides the defining property $\epsilon^i(e_j) = \delta_j^i$ for the elements of the dual basis.

(c) Does the equality in (b) depend on the choice of basis? Conclude how the components of a linear map transform under a change of basis. How, in contrast, do the components of a bilinear map $B: V \times V \rightarrow \mathbb{R}$ transform under the same change of basis?

Solution: No, the equality does not depend on the choice of basis since it has been shown for an arbitrary basis and the associated dual basis in the dual vector space. Since we know that the components $(T_\varphi)^i_j$ transform under a change of basis as the components of a $(1, 1)$ -tensor. Thus we conclude that with respect to new bases

$$\tilde{\epsilon}_k = \bigoplus_{m=1}^{\dim V} C^m_k \odot e_m \quad \text{and} \quad \tilde{\epsilon}^k = \bigoplus_{m=1}^{\dim V} (C^{-1})^k_m \boxminus \epsilon^m,$$

where $\sum_{m=1}^{\dim V} C^m_\ell (C^{-1})^\ell_k = \delta^m_k$ such that still $\tilde{\epsilon}^i(\tilde{e}_j) = \delta^i_j$, the components of the map φ are

$$\tilde{\varphi}^i_j = (\tilde{T}_\varphi)^i_j = T_\varphi(\tilde{\epsilon}^i, \tilde{e}_j) = \sum_{m=1}^{\dim V} \sum_{k=1}^{\dim V} (C^{-1})^i_m C^k_j T_\varphi(\epsilon^m, e_k) = \sum_{m=1}^{\dim V} \sum_{k=1}^{\dim V} (C^{-1})^i_m C^k_j \varphi^m_k.$$

The components of a bilinear map $B : V \times V \rightarrow \mathbb{R}$, in contrast, transform as

$$\tilde{B}_{ij} = B(\tilde{e}_i, \tilde{e}_j) = \sum_{m=1}^{\dim V} \sum_{k=1}^{\dim V} C^a_i C^b_j B(e_a, e_b) = \sum_{m=1}^{\dim V} \sum_{k=1}^{\dim V} C^a_i C^b_j B_{ab}.$$

- (d) Derive that the components $(\psi \circ \varphi)^i_j$ of the composition $\psi \circ \varphi : V \rightarrow V$ of two linear maps $\varphi : V \rightarrow V$ and $\psi : V \rightarrow V$ are given by

$$(\psi \circ \varphi)^i_j = \sum_m^{\dim V} \psi^i_m \varphi^m_j.$$

Explain how this result gives rise to the well-known multiplication rule for matrices.

Solution: Expanding the image of $\psi \circ \varphi$ as in part (b), one has

$$(\psi \circ \varphi)(e_j) = \bigoplus_{i=1}^{\dim V} (\psi \circ \varphi)^i_j \odot e_i,$$

while the left hand side can be written

$$(\psi \circ \varphi)(e_j) = \psi(\varphi(e_j)) = \psi\left(\bigoplus_{m=1}^{\dim V} \varphi^m_j \odot e_m\right) = \bigoplus_{m=1}^{\dim V} \varphi^m_j \odot \psi(e_m) = \bigoplus_{i=1}^{\dim V} \sum_{m=1}^{\dim V} (\varphi^m_j \psi^i_m) \odot e_i,$$

from which the claim follows by comparison with the first equation and commutativity of the multiplication on $\mathbb{R}$. With the convention that the components A^i_j of a linear map

$$\varphi : V \rightarrow V$$

are stored in a $\dim V \times \dim V$ matrix $\mathbf{R}_\varphi$ such that the first index from the left indicates the row while the second index indicates the column, the equation

$$(\psi \circ \varphi)^i_j = \sum_m^{\dim V} \psi^i_m \varphi^m_j$$

can be written as

$$\mathbf{R}_{\psi \circ \varphi} = \mathbf{R}_\psi \cdot \mathbf{R}_\varphi,$$

where the matrix multiplication $\cdot$ on the right hand side is calculated by the familiar rule that proceeds according to the rule “row times column”; indeed, note that on the right hand side of the component formula, the row of the first factor matrix is fixed to i while the column index is running and that the column of the second factor matrix is fixed to j while the row index is running. Thus the standard matrix multiplication procedure implements the composition $\circ$ between maps in terms of the representing matrices $\mathbf{R}_\varphi$ and $\mathbf{R}_\psi$ and yields the representing matrix $\mathbf{R}_{\psi \circ \varphi}$.

- (e) The dual map of a linear map $\varphi : V \rightarrow V$ is the linear map

$$\varphi^* : V^* \rightarrow V^*, \quad \sigma \mapsto \sigma(\varphi(\cdot)),$$

where $\sigma(\varphi(\cdot)) : V \rightarrow \mathbb{R}$ is the linear map defined by $v \mapsto \sigma(\varphi(v))$ and is thus indeed an element of the dual space V^* . Show that expanding the image of the dual map on $(V^*, \boxplus, \boxminus)$,

$$\varphi^*(\sigma) = \bigoplus_{j=1}^{\dim V} \left(\sum_{i=1}^{\dim V} \varphi^*_j{}^i \sigma_i \right) \boxminus \epsilon^j,$$

identifies the components $\varphi^*_j{}^i = \varphi^i_j$. How is the matrix of components for φ^* related to the one for φ ?

Solution: Applying the equation to be shown to the k -th basis vector, one obtains

$$\varphi^*(\sigma)(e_k) = \sum_{j=1}^{\dim V} \sum_{i=1}^{\dim V} \varphi^*_j{}^i \sigma_i \delta_k^j = \sum_{i=1}^{\dim V} \varphi^*_k{}^i \sigma_i.$$

The left hand side of this equation can then be written as

$$\varphi^*(\sigma)(e_k) = \sigma(\varphi(e_k)) = \bigoplus_{i=1}^{\dim V} (\sigma_i \boxtimes \epsilon^i) \left(\bigoplus_{j=1}^{\dim V} \varphi^j{}_k \odot e_j \right) = \sum_{i=1}^{\dim V} \sum_{j=1}^{\dim V} \varphi^j{}_k \sigma_i \delta_j^i = \sum_{i=1}^{\dim V} \varphi^i{}_k \sigma_i,$$

so that comparison of both sides yields $\varphi^*_k{}^i = \varphi^i{}_j$. Thus the representing matrix $\mathbf{R}_{\varphi^*}$ for the dual map φ^* is the transpose of the representing matrix for the map φ ,

$$\mathbf{R}_{\varphi^*} = \mathbf{R}_{\varphi}^T.$$

(f) A linear map $\varphi : V \rightarrow V$ is called *symmetric with respect to an inner product* $g : V \times V \rightarrow \mathbb{R}$ if

$$g(\varphi(v), v') = g(v, \varphi(v')) \quad \text{for all } v, v' \in V.$$

Express this relation entirely in terms of components with respect to some basis and then rewrite it in terms of

$$\varphi_{ij} := \sum_{m=1}^{\dim V} g_{im} \varphi^m{}_j.$$

Simplify the result as much as possible using the properties of inner products.

Solution: Employing a basis on V , one has, on the one hand,

$$g(\varphi(e_i), e_j) = g\left(\bigoplus_{k=1}^{\dim V} \varphi^k{}_i \odot e_k, e_j\right) = \sum_{k=1}^{\dim V} \varphi^k{}_i g(e_k, e_j) = \sum_{k=1}^{\dim V} \varphi^k{}_i g_{kj}$$

and, on the other hand,

$$g(e_i, \varphi(e_j)) = g\left(e_i, \bigoplus_{k=1}^{\dim V} \varphi^k{}_j \odot e_k\right) = \sum_{k=1}^{\dim V} \varphi^k{}_j g(e_i, e_k) = \sum_{k=1}^{\dim V} \varphi^k{}_j g_{ik}.$$

Now using the fact that $g_{kj} = g_{jk}$ and the commutativity of the multiplication in $\mathbb{R}$, combining the above results yields

$$\varphi_{ji} = \varphi_{ij}.$$

Note that this is not identical with the symmetry $\mathbf{R}_{\varphi} = \mathbf{R}_{\varphi}^T$ of the representing matrix for φ . These conditions only coincide in a basis where the components of the tensor g take values $g_{aa} = 1$ for $a = 1, \dots, \dim V$ and vanish for all other index combinations. The usual talk about such an inner product being the ‘standard inner product’ is utterly meaningless, since it is not independent of the choice of basis.